C1.3 Photosynthesis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c1-3.html

Written for this site from the class slides. The chapter is on pages 425–438 of the course textbook.
Image credits are on the last slide of the slides for this subtopic.

C1.3.1 From light energy to chemical energy

pp. 425–438

Photosynthesis transforms **light energy** into **chemical energy**, stored in carbon compounds. This supplies most of the chemical energy needed for life processes in ecosystems.

C1.3.2 Making glucose from carbon dioxide

pp. 425–438

carbon dioxide + water → glucose + oxygen. Water is split, using light energy, and its hydrogen is used to reduce carbon dioxide.

C1.3.3 Oxygen as a by-product

pp. 425–438

Plants, algae and cyanobacteria release oxygen, which comes from the splitting of water. It is a by-product of this process, and it diffuses out.

Watch out: the oxygen comes from water, and not from carbon dioxide.

C1.3.4 Separating pigments by chromatography

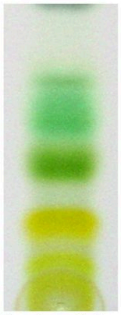
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A leaf extract is spotted on a strip, and a solvent moves up it. Pigments separate, as they differ in solubility.

Rf = distance moved by pigment / distance moved by solvent. Identify pigments by colour and by Rf. For example, a pigment moving 36 mm with a solvent moving 80 mm has Rf = 0.45.

Separating leaf pigments by chromatography

C1.3.4



Pigments on a strip. Flo, CC BY-SA 2.5.

1. Grind a leaf with solvent to extract pigments.
2. Spot the extract on a **pencil line**.
3. Stand the strip in solvent, spot above the surface.
4. Pigments move different distances because they differ in solubility.
5. Mark the solvent front.

Pigment	Colour
Carotene	Orange-yellow
Xanthophyll	Yellow
Chlorophyll a	Blue-green
Chlorophyll b	Yellow-green

Safety
Ventilated room, no flames, eye protection.

Why pencil?
Ink would dissolve and separate too.

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C1.3.5 How pigments absorb light

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When a pigment absorbs a photon, an **electron is excited** to a higher energy level, so light energy has become chemical energy. Only some wavelengths have the right energy to do this. Chlorophyll absorbs red and blue best, and reflects green.

C1.3.6 Absorption and action spectra

pp. 425–438

	Absorption spectrum	Action spectrum
Vertical axis	Percentage of light absorbed	Rate of photosynthesis
Similarities	Both peak in red and blue, and both are lower in green	
Differences	Rate does not fall as low as absorption in green, as accessory pigments and scattering help	

C1.3.7 Investigating limiting factors

pp. 425–438

The rate is limited by whichever factor is furthest from its optimum. To investigate: move a lamp to vary **light intensity**, add sodium hydrogencarbonate to vary **CO_2 concentration**, and use water baths to vary **temperature**. Measure oxygen production, or carbon dioxide use by following pH.

C1.3.8 CO_2 enrichment experiments

pp. 425–438

Enclosed greenhouses add CO_2 to a closed space, which makes it easier to control other factors. **FACE** experiments release CO_2 over open plots, so conditions are closer to natural ecosystems. Growth often rises at first, but by less than predicted.

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